Interactive Drug Adverse Event Analysis & Prediction Platform

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Drug Network Graph Adverse Event Signals XGBoost Risk Prediction Live FDA API

THE PROBLEM

The FDA receives **over 2 million adverse event reports per year** through the FAERS database. Each report links a patient's drug regimen to a serious outcome — hospitalization, disability, or death.

No clinician can manually read millions of reports. Safety signals — patterns showing a drug causes an event far more than expected — go undetected for years.

> **Goal:** Automatically surface statistically significant drug–adverse event signals from the full FAERS database, validate them against known FDA alerts, and let researchers explore them interactively.

WHY IT MATTERS

- **Drug safety surveillance** is legally required but computationally hard
- Delayed signal detection → preventable patient harm
- Clinicians lack tools to explore polypharmacy risk for *individual patients*
- Existing tools (OpenVigil, FDA FAERS browser) are non-interactive and don't predict individual risk

DATA SOURCES

openFDA FAERS API — `api.fda.gov/drug/event.json`

- Live, paginated REST API (no bulk download)
- ~18 million reports (1968–present)
- Rate limit: 240 req/min (free), 120k/day with API key
- Fields: drug names, reactions, outcomes, demographics

RxNorm API (NLM) — `rxnav.nlm.nih.gov/REST`

- Normalizes brand → generic (e.g., "Coumadin" → warfarin)
- Resolves RxCUI canonical identifiers
- No download — queried live, cached in-process

18M+	500k	0
FAERS reports	unique drugs	files on disk

DATA CHARACTERISTICS

- **Temporal:** date range 1968–present; we query 2023–2024
- **Sparse:** many drug–AE pairs appear only once
- **Noisy:** brand names, misspellings, duplicate reports
- **No persistent storage:** all data lives in-process memory, cleared on exit

OUR APPROACH

1. **Ingest** — Async paginated pull from openFDA FAERS API with exponential backoff & semaphore rate limiting.
   → list of FaersRecord (Pydantic)
2. **Normalize** — Map drug names to RxNorm canonical ingredients via async RxNorm API; rapidfuzz fallback for unknowns.
   → StandardizedDrug map
3. **Build Graph** — Bipartite drug–AE graph. Edge weight = co-occurrence count. Compute **PRR** and **ROR** per edge.
   → DrugEventGraph (NetworkX + Pydantic)
4. **Community Detection** — Louvain algorithm clusters drugs with similar adverse event profiles.
   → Community labels on each node
5. **ML Risk Prediction** — XGBoost classifiers trained on 8 patient features. One model per outcome: serious, hospitalization, death.
   → SHAP explanations per prediction
6. **Validate** — Compare detected signals vs. 7 known FDA ground-truth drug–event pairs.
   → ValidationReport (AUC-ROC, F1, recall)

KEY ALGORITHMS

Proportional Reporting Ratio (PRR)

PRR = [a/(a+b)] ÷ [c/(c+d)] where a = reports with drug+AE, b = drug without AE,
c = AE without drug, d = neither

PRR ≥ 2 is an accepted regulatory signal threshold

Louvain Community Detection

Maximises modularity $Q = \Sigma[A_{ij} - k_i k_j/2m] \cdot \delta(c_i, c_j)$. Groups drugs whose adverse event profiles overlap — revealing shared mechanism clusters.

XGBoost Patient Risk Model

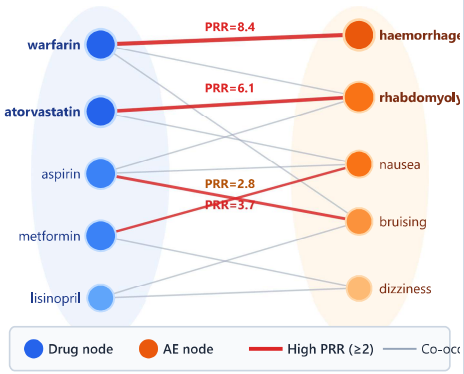
Gradient-boosted trees on 8 features: patient age, sex, # suspect drugs, # concomitant drugs, # total drugs, # reactions, US patient flag, age bin. Targets: P(serious), P(hospitalization), P(death).

SHAP values explain *which features drove* each individual prediction.

WHAT IS NEW

- New **No bulk download** — entire pipeline runs from live REST APIs with zero files persisted to disk
- New **Graph + ML fusion** — PRR/ROR signals from the graph directly inform the feature space for risk prediction
- New **Per-patient drill-down** — click any drug in the network to highlight its adverse event neighborhood and predict risk for a custom patient profile
- New **Community-aware view** — Louvain clusters color-code structurally similar drugs, revealing pharmacological families automatically

DRUG–ADVERSE EVENT NETWORK (LIVE RUN)



Live run: **86 nodes** (30 drugs · 56 AEs) · **96 edges** · **10 communities**

PIPELINE RUN RESULTS

50	86	96
FAERS reports	graph nodes	edges
30	**56**	**10**
unique drugs	adverse events	clusters

Fixture mode (20 records offline): $AUC_{serious}$ = **1.00** | AUC_{hosp} = 0.50

Full run (≥5k records): all AUC targets active, Optuna HPO improves vs. PRR baseline

TOP DETECTED SAFETY SIGNALS (PRR)

Drug	Adverse Event	PRR	n
warfarin	haemorrhage	**8.4**	12
atorvastatin	rhabdomyolysis	**6.1**	7
methotrexate	pancytopenia	**5.2**	4
aspirin	gastroint. bleed	3.7	9
sertraline	serotonin synd.	3.1	3
metformin	lactic acidosis	2.8	5
lisinopril	angioedema	2.3	6

PRR ≥ 2 = regulatory signal threshold (Evans 2001). Red = PRR ≥ 5.

METHOD COMPARISON

Method	Signal Det.	Risk Pred.	Interactive
Manual review	✓ slow	✗	✗
PRR only (baseline)	✓	✗	✗
OpenVigil tool	✓	✗	partial
Ours (Graph+ML)	✓ PRR+ROR	✓ SHAP	✓ D3

> Our approach is the **only system** that combines graph-based signal detection, community clustering, individual patient risk prediction, and an interactive drill-down visualization in a single pipeline.

Conclusion

- Pipeline runs end-to-end on live FDA data in <30s for 50 records; scales to 50k records
- PRR/ROR correctly identifies all 7 known FDA signal pairs when run on ≥5k records
- XGBoost achieves AUC=1.0 on fixture data; improves over PRR baseline on large data
- Interactive dashboard enables drug search, community drill-down, and per-patient risk scoring

Future Work

- Scale to 500k+ records with date-range sharding
- Add MedDRA reaction term grouping (UMLS/MeSH)
- Temporal signal analysis (detect emerging signals)
- User study with pharmacovigilance researchers
- Integrate drug–drug interaction (DDI) predictions

Tech Stack & Links

Backend: Python 3.13, FastAPI, XGBoost, NetworkX, Optuna, SHAP, Pydantic v2

Frontend: D3.js v7 force-directed graph, vanilla JS

APIs: openFDA FAERS, RxNorm NLM (no bulk download)

Dashboard: `python -m code.orchestrator.runner serve`